1 **(a)** $2 \times (12 + 7)$ **[M1]**

38 **[A1]**

(b) 12×5 **[M1]**

60 **[A1]**

[Total Marks: 4]

2 **(a)** Calculates area of triangle: $\frac{1}{2} \times 12 \times 8 = 48$ **[M1]**

$4x = 48$

x = 12[$A1$]

(b) Uses Pythagoras to find slanted side: $\sqrt{6^2 + 8^2}$ **[M1]**

Slanted side = 10 **[A1]**

Perimeter $= 12 + 10 + 10 = 32$ **[A1]**

[Total Marks: 5]

3 **(a)** $15 \times h = 120$ **[M1]**

8 **[A1]**

(b) $2(15 + AD) = 54$ OR $15 + AD = 27$ **[M1]**

12 **[A1]**

(c) 60 **[B1]**

[Total Marks: 5]

4 **(a)** $2(2x + 3) + 2(x - 1) = 34$ OR $2x + 3 + x - 1 = 17$ **[M1]**

$6x + 4 = 34$ OR $3x + 2 = 17$ **[M1]**

$x = 5$ **[A1]**

(b) Substitution of $x = 5$: $(2(5) + 3) \times (5 - 1)$ OR 13×4 **[M1]**

52 **[A1]**

[Total Marks: 5]

5 **(a)** $\frac{1}{2}(10 + 16) \times 8$ **[M1]**

104 **[A1]**

(b) Base of right-angled triangle $= 16 - 10 = 6$ **[M1]**

Slanted edge $= \sqrt{6^2 + 8^2} = 10$ **[M1]**

$8 + 10 + 16 + 10 = 44$ **[A1]**

[Total Marks: 5]

6 **(a)** $\frac{1}{2}(9+15)\times 8$ **[M1]**

96 **[A1]**

(b) $BC^2 = (15-9)^2 + 8^2 = 36 + 64 = 100$ **[M1]**

$BC = 10$ cm **[A1]**

Perimeter $= 9 + 10 + 15 + 8 = 42$ cm **[B1]**

(c) $DE = 15 - 9 = 6$ cm and area $= \frac{1}{2} \times 6 \times 8$ **[M1]**

24 **[A1]**

[Total Marks: 7]

7 **(a)** $2((3x+2)+(x+4))$ or $2(4x+6)$ **[M1]**

$8x + 12$ **[A1]**

(b) Area of rectangle $PQRS = (3x+2)(x+4) = 3x^2 + 12x + 2x + 8 = 3x^2 + 14x + 8$ **[M1]**

Area of parallelogram $ABCD = x(2x-1) = 2x^2 - x$ **[B1]**

$A = (3x^2 + 14x + 8) - (2x^2 - x) = x^2 + 15x + 8$ **[A1]**

(c) **(i)** $x^2 + 15x + 8 = 62$ giving $x^2 + 15x - 54 = 0$ **[M1]**

$(x+18)(x-3) = 0$ **[M1]**

3 (rejecting $x = -18$ as lengths must be positive) **[A1]**

(ii) $3(2(3) - 1) = 15$ **[B1]**

[Total Marks: 9]

8 **(a)** Uses Area $= \frac{1}{2} \times$ base $\times$ height $= \frac{1}{2} \times 14 \times 9$ **[M1]**

63 **[A1]**

(b) Finds $AE = \sqrt{9^2 + 9^2} = \sqrt{162}$ or $12.7\ldots$ **[M1]**

Finds $BE = \sqrt{5^2 + 9^2} = \sqrt{106}$ or $10.2\ldots$ **[M1]**

Perimeter $= 14 + \sqrt{162} + \sqrt{106}$ **[M1]**

37.0 (or $37.02\ldots$) **[A1]**

[Total Marks: 6]

9 **(a)** $\frac{1}{2} \times (15+24) \times 8$ **[M1]**

156 **[A1]**

(b) Horizontal distance from D to the altitude from $A = \sqrt{10^2 - 8^2} = 6$ **[M1]**

Horizontal distance for the right-angled triangle under $B = 24 - 15 - 6 = 3$ **[M1]**

$BC = \sqrt{8^2 + 3^2}$ $(= \sqrt{73} \approx 8.544)$ **[M1]**

57.5 OR 57.54 to 57.55 **[A1]**

[Total Marks: 6]

10 **(a)** $\frac{1}{2} \times (10 + 22) \times 8$ **[M1]**

128 **[A1]**

(b) Finds horizontal base projection of slanted side: $\frac{22 - 10}{2} = 6$ **[B1]**

Applies Pythagoras' theorem: $PS = \sqrt{6^2 + 8^2}$ **[M1]**

$PS = 10$ **[A1]**

Perimeter $= 10 + 22 + 10 + 10 = 52$ **[A1]**

[Total Marks: 6]

11 **(a)** Area of rectangle $= 4(2x + 3)$ OR $8x + 12$ **[M1]**

Area of triangle $= \frac{1}{2} \times 6 \times (3x - 1)$ OR $9x - 3$ **[M1]**

$8x + 12 = 9x - 3$ **[M1]**

$x = 15$ **[A1]**

(b) Length $= 2(15) + 3 = 33$ **[M1]**

Perimeter $= 2(4 + 33) = 74$ **[A1]**

[Total Marks: 6]

12 **(a)** $\frac{1}{2} \times 8 \times ((x + 4) + (3x - 2)) = 120$ **[M1]**

$4(4x + 2) = 120$ OR $4x + 2 = 30$ **[M1]**

7 **[A1]**

(b) Evaluates parallel sides: 11 and 19 **[B1]**

Uses Pythagoras' theorem to find the slanted side: $\sqrt{8^2 + \left(\frac{19 - 11}{2}\right)^2}$ OR $\sqrt{8^2 + 4^2}$ [M2]

(M1 for finding the horizontal base of 4)

47.9 (47.88 . . .) **[A1]**

[Total Marks: 7]

13 **(a)** **(i)** Sets up equation for area: $\frac{1}{2}((x + 4) + (3x - 2))(x - 1) = 20$ **[M1]**

Simplifies expression to $\frac{1}{2}(4x + 2)(x - 1) = 20$ or $(2x + 1)(x - 1) = 20$ **[M1]**

Expands to $2x^2 - 2x + x - 1 = 20$ leading to $2x^2 - x - 21 = 0$ with no errors seen **[A1]**

(ii) Factorises $(2x - 7)(x + 3) = 0$ or uses quadratic formula to find $x = 3.5$ **[M1]**

Calculates straight sides: $AD = 7.5$, $BC = 8.5$, $AB = 2.5$ **[M1]**

Sets up Pythagoras for the slanted side CD: $\sqrt{(8.5-7.5)^2+2.5^2}$ **[M1]**

Calculates $CD = \sqrt{1^2+2.5^2} = \sqrt{7.25}$ (or $2.69\ldots$) **[A1]**

Perimeter $= 7.5 + 8.5 + 2.5 + 2.69\cdots = 21.2$ (or $21.19\ldots$) **[A1]**

[Total Marks: 8]

14 (a) Equates areas: $\frac{1}{2}(2x+1)(3x-2) = (x+4)(2x-1)$ **[M1]**

Expands brackets: $\frac{1}{2}(6x^2 - x - 2) = 2x^2 + 7x - 4$ **[M1]**

Multiplies by 2 and simplifies to $6x^2 - x - 2 = 4x^2 + 14x - 8$, leading to $2x^2 - 15x + 6 = 0$ **[A1]**

(b) Uses quadratic formula: $x = \dfrac{-(-15) \pm \sqrt{(-15)^2 - 4(2)(6)}}{2(2)}$ **[M1]**

$x = \dfrac{15 \pm \sqrt{177}}{4}$ **[M1]**

7.08 and 0.42 **[A1]**

(c) Selects valid root $x = 7.076...$ (since $x = 0.42$ gives $EF < 0$ and $BC < 0$) and substitutes into $2(3x+3)$ **[M1]**

48.5 (accept 48.46 to 48.5) **[A1]**

[Total Marks: 8]

15 (a) $\frac{1}{2}(x + x + 4)(x-1) = 40$ **[M1]**

$\frac{1}{2}(2x+4)(x-1) = 40$ OR $(x+2)(x-1) = 40$ **[M1]**

$x^2 + 2x - x - 2 = 40 \implies x^2 + x - 42 = 0$ **[A1]**

(b) $(x+7)(x-6) = 0$ **[M1]**

6 **[A1]**

(c) Identifies lengths are Top = 6, Bottom = 10, Height = 5 **[M1]**

Horizontal difference = 4, so slanted side $= \sqrt{5^2+4^2}$ OR $\sqrt{41}$ **[M1]**

Perimeter $= 5 + 6 + 10 + \sqrt{41}$ **[M1]**

27.4 (or $27.40\ldots$) **[A1]**

[Total Marks: 9]

16 (a) Area of rectangle $= (2x+1)(x+3) = 2x^2 + 6x + x + 3$ **[M1]**

Area of triangle $= \frac{1}{2}(3x+2)(2x-1) = \frac{1}{2}(6x^2 - 3x + 4x - 2)$ **[M1]**

Equates areas: $2x^2 + 7x + 3 = \frac{1}{2}(6x^2 + x - 2)$ **[M1]**

Multiplies by 2 and simplifies correctly to $2x^2 - 13x - 8 = 0$ with no errors **[A1]**

(b) $\dfrac{-(-13) \pm \sqrt{(-13)^2 - 4(2)(-8)}}{2(2)}$ [M2]

(M1 for correct substitution into quadratic formula)

$\dfrac{13 \pm \sqrt{233}}{4}$ **[M1]**

7.07 and -0.57 **[A1]**

(c) Substitutes the positive value of x into $2(2x + 1 + x + 3)$ OR calculates sides 15.14 and 10.07 **[M1]**

50.4 (accept 50.39 to 50.43) **[A1]**

[Total Marks: 10]

17 (a) Perimeter of rectangle $= 6y$ OR Perimeter of square $= 60 - 6y$ **[M1]**

$\dfrac{60 - 6y}{4}$ OR $15 - 1.5y$ **[A1]**

(b) $(15 - 1.5y)^2 + y(2y) = 113$ **[M1]**

$225 - 45y + 2.25y^2 + 2y^2 = 113$ **[M1]**

$4.25y^2 - 45y + 112 = 0$ leading to $17y^2 - 180y + 448 = 0$ with no errors seen **[A1]**

(c) $\dfrac{-(-180) \pm \sqrt{(-180)^2 - 4(17)(448)}}{2(17)}$ **[M1]**

$\dfrac{180 \pm \sqrt{1936}}{34}$ OR $\dfrac{180 \pm 44}{34}$ **[M1]**

$y = 4$ and $y = 6.59$ (or 6.588... or $\dfrac{112}{17}$) **[A1]**

(d) Identifies $y = 4$ as the valid solution (since $15 - 1.5(6.59) < 6.59$) leading to side length $= 9$ **[M1]**

81 **[A1]**

[Total Marks: 10]

18 (a) $BD^2 = 14.5^2 + 9.2^2 - 2(14.5)(9.2)\cos 118°$ **[M1]**

$BD^2 = 210.25 + 84.64 - (-125.258\ldots) = 420.148\ldots$ **[M1]**

20.5 **[A1]**

(b) $\dfrac{\sin \angle ABD}{9.2} = \dfrac{\sin 118°}{20.497\ldots}$ giving angle $ABD = 23.347\ldots°$ **[M1]**

Angle BDC = angle $ABD = 23.3°$ (alternate angles, AB is parallel to DC) **[A1]**

(c) $DC = 20.497\ldots \cos 23.347\ldots° = 18.819\ldots$ cm **[M1]**

$BC = 20.497\ldots \sin 23.347\ldots° = 8.123\ldots$ cm **[M1]**

Perimeter $= 14.5 + 8.123\cdots + 18.819\cdots + 9.2 = 50.6$ **[A1]**

(d) $\dfrac{1}{2}(14.5 + 18.819\ldots) \times 8.123\ldots$ (or $\dfrac{1}{2}(14.5)(9.2)\sin 118° + \dfrac{1}{2}(18.819\ldots)(8.123\ldots)$) **[M1]**

135 **[A1]**

[Total Marks: 10]